Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 12 Handout

Thursday, October 22

Problem 12.1

Suppose we want to build a classifier that labels email as spam or not spam based on whether certain words are present in the email, based on a somewhat simplified probabilistic model. For a given email, let indicator random variable C be equal to 1 if the email is spam and 0 if the email is not spam, and $s \triangleq p_C(1)$. We have a dictionary of n words and use an indicator random variable Y_i to capture whether word i from the dictionary appears in the email ($Y_i = 1$) or not ($Y_i = 0$). We assume that $Y_1, \ldots, Y_n$ are conditionally independent given C[1], and $p_i \triangleq p_{Y_i|C}(1|1)$, $q_i \triangleq p_{Y_i|C}(1|0)$, $i = 1, \ldots, n$. Assume that parameters $s, p_1, \ldots, p_n, q_1, \ldots, q_n$ are unknown. To build the classifier, we will estimate these parameters from a training set of k emails. Email j in the training set has a label $c^{(j)}$, which tells us whether the email is spam or not. For email j, we also generate n features $y^{(j)} = \left(y_1^{(j)}, y_2^{(j)}, \ldots, y_n^{(j)}\right)$ where $y_i^{(j)}$ is the observed value of Y_i for this email. We shall assume that different emails are generated i.i.d.from the same distribution.

(a) First, we estimate the parameters $\theta = (s, p_1, p_2, \ldots, p_n, q_1, q_2, \ldots, q_n)$.

(i) As a warm-up before doing any calculations, what is a reasonable way to estimate p_i and q_i?

(ii) Determine functions $f, g_1, \ldots, g_n, h_1, \ldots, h_n$ such that

$$\ell\left(\theta; c^{(1,\ldots,k)}, y_{1,\ldots,n}^{(1,\ldots,k)}\right) \triangleq \log\left(\prod_{j=1}^{k} p_{C,Y_1,\ldots,Y_n}(c^{(j)}, y_1^{(j)}, \ldots, y_n^{(j)}; \theta)\right)$$
$$= f(s) + \sum_{i=1}^{n} g_i(p_i) + \sum_{i=1}^{n} h_i(q_i).$$

Your answer may depend on the training data $\left\{c^{(1,\ldots,k)}, y_{1,\ldots,n}^{(1,\ldots,k)}\right\}$.

(iii) Determine the ML estimates $\hat{s}, \hat{p}_1, \ldots, \hat{p}_n, \hat{q}_1, \ldots, \hat{q}_n$ of the model parameters.

(b) Once we estimate the model parameters, we can treat them as fixed and classify new emails as spam or not spam.

[1]This assumption of the model is not actually true for emails since certain words may be more likely to co-occur. Also, the model does not account for the ordering of the words or whether a word occurs multiple times.

(i) Suppose the new email has features $Y_1 = y_1, \ldots, Y_n = y_n$. Determine the probability that $C = 1$ (the email is spam).

(ii) Suppose a word appears in the new email (e.g., $Y_1 = 1$) but did not appear in the training data (i.e., $Y_1^{(j)} = 0$ for $j = 1, \ldots, k$). Compute $p_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n)$. What went wrong?

(iii) The result of part (ii) suggests that ML estimates can lead to a model that is fragile. Suggest a way to fix this problem.